Malachi Mabie

AI/ML Research & Systems Engineer

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in *mlmabie*

Applied research, physical systems, cognitive security, graph/vision ML, agent infrastructure, and production software.

Summary

Engineer and applied AI researcher with roots in FIRST Robotics, physics, music, robotic tooling, medical-device manufacturing, audio tiny-ML, chip-test instrumentation, cognitive science, and neural data science. Later work spans defense AI/ML R&D, cognitive security, graph/vision ML, agent infrastructure, production startup software, and robotic welding research. Finds the computationally tractable edge of poorly specified domains, then builds the representations, data, evaluation loops, hard validators, and software systems that move it.

Experience

AI Software Engineer / Independent Contractor, Mabie Industries, LLC, Carpinteria, CA (remote) *Apr 2025–Present*

Independent engineering practice spanning robotics, construction AI, cognitive security, decentralized consumer AI, e-commerce, GIS, and B2C software. Individual engagements below.

- Built an agent research runtime used across engagements: MCP orchestration, typed tool contracts, dynamic subagents, multi-store memory, retrieval, verifier scoring, run records, regression gates, and DSPy/GEPA optimization against recorded scores over a large graph-linked research corpus
- Recurring method across clients: preserve the source artifact, stabilize object identity, expose typed relations, keep hard checks outside the learned stack, and spend model capacity only on residual ambiguity

Robotics ML Research Engineer (contract), Advanced Metal Research (Gradient Industrial Robotics Corporation), Hawthorne, CA (hybrid) *Jul 2026–Present*

- Lead research software engineering across the robotic-welding stack: weld and materials data, learning and controls algorithms, motion planning, perception, multi-factor optimization, harness ergonomics, evaluation, and process-off integration
- Defined typed weld-book, material, joint, process-window, study, trajectory, metallurgy-output, trace, and evidence models; added randomized experiments and content-addressed capture of human and simulator trajectories
- Treated torch orientation and robot/positioner coordination as optimization variables rather than fixed constraints, with independent kinematic, collision, calibration, continuity, and process checks held authoritative over execution
- Bound replayable run evidence to source revision, binaries, robot model, calibration, collision assets, and request/response pairs, so a planning result can be re-derived rather than trusted
- Designed the staged path from simulation to supervised process-off trials, including the data contracts for measured motion, operator corrections, and sensor residuals that later learning experiments consume
- Built Steam Deck operator controls and robot runtime integration; resolved client/runtime packet, identity, and configured-axis mismatches, contributing to operator-confirmed jogging on the assembled robot. Full trajectory execution and welding remain separate validation work

ML & Product Cognitive Security (contract), Siethar Group, Remote *2025–2026*

- Cognitive-security engineering for AI-mediated decision surfaces: product threat models, human/agent/tool trust boundaries, and agent/human workflow analysis
- Adversarial training and evaluation for failure modes that appear only in AI-native interaction, built as concrete failure-mode literacy rather than demo material

Construction Geometry ML / DXF-to-CAD (contract & independent), Augrade-facing research, Remote *2026*

- Vector-native CAD tokenization, provenance-preserving supervectors, HATCH-IoU self-supervision, typed constraints, rewrite-invariant merge logic, hard validators outside the learned stack, and learned repair/search for residual ambiguity
- Public workbench in stdlib Python (`github.com/mlmabie/DXF_Primitive_to_Polygon`): recovered 2,237 wall, column, and curtain-wall polygons from roughly 67,000 primitives across 111 layers via planar-graph face-walking, with parameter sweeps and SVG review artifacts
- Design rule: preserve variance until the system can distinguish a rewrite from a provenance signal from a semantic distinction from noise
- Frames GNNs as one layer inside a legible geometry/constraint stack rather than as the whole system

AI Operations & Analytics (contract), Hammies, Remote *2025–2026*

- Converted ambiguous growth and customer/product questions into lightweight automations, structured analytics, and usable operating loops for product marketing

Software Engineer (contract), Drone Institute, Remote *May–Dec 2025*

- Six production systems end to end: thermal-imagery/GIS workflows, multi-tenant marketplace, partner onboarding portal, operations dashboard, webhook-driven order flow, and cloud deployment
- Event-driven serverless architecture with durable state and stable identity boundaries; ran at low fixed monthly cost against $811k ARR potential
- Owned the full lifecycle from ambiguous requirements and workflow discovery through architecture, implementation, deployment, production debugging, analytics, and operator/customer experience
- Also delivered cloud-architecture consultation, technical specifications, and development-standards recommendations

Visiting Researcher (contract), Truffle Computer Company (DeepShard Inc.), Los Angeles, CA *Apr 2025*

- Multi-agent protocols, tool calling, agentic AI tooling, networking and memory systems, LLM optimization, product design, and software ergonomics
- Wrote an architecture comparison of the Prime Intellect and Gensyn approaches to distributed RL and compute; brought Named Data Networking and defense-AI perspective into consumer AI

RAM Laboratories, Inc., Full-time, 3 years, San Diego, CA (hybrid/remote) *Apr 2022–Mar 2025*

Senior AI Research Engineer (Apr 2024–Mar 2025) · AI/ML Software Research Engineer (Apr 2022–Apr 2024). Internally Member of Technical Staff → Senior MTS → Principal Investigator.

- Principal Investigator on T-UEBA, a $250k Army SBIR in tactical user/entity behavior analytics: constrained heterogeneous temporal GNN over user-resource-action graphs with calibrated graph attention, Hopfield conditioning, conformal prediction, sparse-autoencoder interpretability, active learning, and curriculum training
- T-UEBA results: 95% accuracy, 0.887 F1, <3% false-positive rate, 15.2ms/graph, 15MB model — 400× smaller and 7.9× faster than baseline, under shifting data and hard compute limits
- T-UEBA rule discovery: because the model was built to be interpretable, it generated stronger access-control rule sets for the canonical Elastic Stack zero-trust policy decision point than those in use, so the customer could run the GNN inline or adopt the improved rules on their existing PDP
- Invented DL-Patcher for an MDA-sponsored SBIR: automated C/C++ vulnerability repair with synthetic bug insertion, natural-patch filtering on function-level change statistics and similarity, static-checker feedback, patch verification, and a VS Code human-in-loop workflow. Phase I seq2seq models across four memory-related bug types; Phase II RoBERTa work; U.S. patent filed
- Directed VULCAN, RAM's DARPA AI Cyber Challenge (AIxCC) submission: code-model harnesses across RoBERTa, GraphCodeBERT, CodeT5++, and LLaMA QLoRA, with static analysis, decompilation, fuzzing, and runtime verification supplying accept/reject signal on generated patches. Tracked separately from the DL-Patcher lineage
- Drove COG-SAFE under DARPA FACT: architecture for accountable transactions among humans, agents, and tools with explicit trust boundaries, behavior monitoring, and adversarial robustness; built with cognitive-psychology researchers and a Georgia Tech collaboration; represented RAM in Google foundation-model meetings
- Owned the data and objective side across programs: ingestion and filtering, train/eval splits, synthetic bug insertion, natural-patch mining, active learning on uncertain examples, and curriculum and data-centric evaluation. The through-line was predictive data debugging — interpreting a model in order to interpret its dataset, so that what the model computes on an example predicts what training on it will move
- Built model-agnostic distributed edge inference: Fused Tile Partitioning, custom PyTorch backend hooks into a C++ runtime, heterogeneous edge execution, and work-stealing scheduling with Named Data Networking content-centric model routing
- Architected containerized tactical-network simulation for adversarial multi-agent coordination: PBFT, HotStuff, and Streamlet consensus with sharding and reputation-based schemes over EMANE tactical networks
- Authored 15+ SBIRs and technical proposals — LEAR, HopModKG, Warfighter AI, VAAST, HAYSTAC, DEMMV, distributed edge inference, and tactical-network simulation — converting a literature moving faster than the funding cycle into fundable, executable programs
- Mentored junior researchers and translated fast-moving AI literature into architecture, evaluation strategy, technical narratives, and customer-facing proposals

Machine Learning & AI Instructor, Inspirit AI, Palo Alto, CA (remote) *May 2021–Apr 2022*

Also Deep Dives Instructor (Jul 2021–Apr 2022) and Curriculum Development Manager (Jul–Aug 2021).

- Taught reinforcement learning and deep Q-learning, transformers, BERT for NLP, LSTMs, GANs, model deployment, self-driving, medical imaging, object detection, segmentation, and tracking
- Delivered live workshops, spotlight talks, and capstone mentoring; built Jupyter-notebook curriculum
- Piloted a middle-school AI curriculum and rescued an experimental remote-in-person class at Pilgrim School; mentored Muirlands FLL robotics curriculum from scratch

Neural Data Science Special Project, UC San Diego, Bradley Voytek course, San Diego, CA *2021*

- Proposed and executed a human cortical organoid study combining RNA-seq gene features with multi-electrode-array local-field-potential recordings
- Extracted mean firing rate, Fano factor, interspike interval, center frequency, and percent-electrode-with-oscillation features; used PCA for spike sorting and exploratory analysis
- Built normalized trend-similarity heatmaps between gene-expression and LFP features despite sparse RNA-seq time points; identified exploratory correlations between oscillatory-electrode percentage and glutamatergic/mitotic cell-count trends. Result forwarded to the Muotri lab that recorded the data

Software Engineer Intern, Endura Technologies, San Diego, CA *Oct 2019–Sep 2020*

- Built a Java instrumentation-control platform for SoC chip testing, translating register maps, data sheets, and spreadsheet-defined register functions into usable lab software across varying development stages and user audiences
- Wrote Python code-generation and automation routines with Verilog and EE collaborators, cutting slow manual test-code changes and weekend-scale runtime-testing loops
- Automated register-map translation and register flagging during SoC tests

Deep Learning Research Intern, Endura Technologies, San Diego, CA *Jul–Dec 2019*

- Quantized windowed-audio CNN pipelines with heavy data mining, cleaning, and signal-transform work for tiny-ML deployment on custom AI-accelerator ASICs
- Re-represented audio training snippets into visual/CNN-friendly forms for an application whose deployed algorithm was transform-based
- Connected neural-signal-processing coursework to practical data-quality, representation, and constrained-inference questions in a hardware-near startup

Computer Vision R&D Intern, RoadReader, La Jolla, CA *Jul–Aug 2019*

- Sparsely supervised learning pipelines for sensor fusion and object detection, applied to pothole detection using city vehicles and devices as a distributed sensing platform

Design Researcher, UCSD Design Lab, Automation Playground, San Diego, CA *Oct 2018–Jun 2019*

- Hyundai-sponsored robotaxi and micro-mobility speculative design project for autonomous vehicles; Ford-sponsored contract-mobility design research
- Design research, storyboarding, and speculative design grounded in people-centered design and distributed cognition — the formal introduction to human/AI interaction that later informed cognitive-security work

Polymer Design & Mechanical Engineer, Applied Medical, Lake Forest, CA *Jun 2015–Sep 2018*

- Designed creative fixtures, end-of-arm tooling (EOAT), pneumatically powered robotic tools, manipulators, grippers, and linkages around polymer and metal injection molding, insert molding, and product subassemblies. Mentored by Ron Surya
- Solved a metal-injection insert-molding variance problem with a 3D-printed orientation grid, fishing-line sorting geometry, and stainless suction-pen tips; production implementation saved roughly $100k/month in operating variance and improved shop-order speed
- Concept through CAD feasibility, prototyping, testing, qualification, ANSI Y14.5 detailed drawings, BOMs, assembly drawings, and shop-floor troubleshooting
- Hands-on assembly, wiring, and machine programming; supervised builds of automated and electro-mechanical equipment; worked directly with production staff under medical-device documentation rigor

Process Development Engineer Intern, Applied Medical, Rancho Santa Margarita, CA *Jun 2013–May 2015*

- Started at 16 in polymer process development; concept-to-assembly work on several robotic manipulators used in product manufacturing, via CAD and prototyping
- Hands-on process improvement and maintenance of robotic tooling around injection molding, inside the documentation rigor of a medical-technologies company

Design Co-Lead & Mentor, FIRST Robotics, Teams 1425 & 3476, Orange County, CA *2007–2013*

- Six years across FLL, FTC, and FRC; the mechanical-design foundation that led directly to the Applied Medical internship at 16
- 2013 design co-lead for intake, drive, superstructure, and integration across 500+ parts and assemblies; introduced CAD discipline, a 10-step design process, and rapid prototyping loops
- Mentored FLL students, taught NXT-G programming and manipulator design, refereed and judged events, ran demos, and twice built robotics summer-camp curricula from scratch

Patents

Automated bug fixing using deep learning — U.S. patent application 18/375,839 (filed). Deep-learning pretraining and fine-tuning for automated C/C++ vulnerability repair, with training-feedback algorithms, natural-patch quality filtering, static-checker feedback, and patch verification. Basis of DL-Patcher.

Selected Independent Work

- **Agent/eval harness** (in progress) — minimal harness for tool-using LLM agents: domain-shaped tasks, explicit tool contracts with refusal/uncertainty behavior, replayable trace schema, verifier layer, memory/state perturbation, failure taxonomy, and score reports
- **Cognitive-security game** (in progress) — adversarial training and evaluation for how humans and agents get steered under AI-native interaction
- **Custom Flash Attention** — implemented the week the method was announced; alongside FSDP and SLURM/HPC multi-GPU training work

Education & Training

B.S. Cognitive Science: Machine Learning & Neural Computation, UC San Diego *2021*

Reinforcement Learning, Advanced Machine Learning, Neural Signal Processing, Neural Data Science, Social Neuroscience, Convex Optimization, Probability & Statistics, Data Structures & Algorithms, Systems Programming, HPC.

Physics, Irvine Valley College *2015–2017*

Completed while working in manufacturing design at Applied Medical; physics as the mathematical bridge from mechanical intuition to cognition, computation, signal processing, and AI. Studied music in parallel.

Production-Ready Agent Engineering (MCP → RL), Maven, taught by Will Brown and Kyle Corbitt *2025*

NeurIPS, Attended *2025*

Awards & Recognition

Robotics: FRC Newton Division World Champion; Indiana Robotics Invitational Champion; Industrial Design and Innovation in Control awards; San Diego Regional Champion; Excellence in Engineering Award.

Music: OMTA Gold Cup pianist; 2× OMTA Superior Award; ICM Division III South Contemporary recognition; independent piano teacher at 14; nursing-home performance volunteer.

Design & creative: UCSD Designathon winner; National Geographic Kids photography publication; oil painting and landscape/abstract photography.

Technical Surface

Research & Models
GNNs, Transformers, EBMs
Hopfield Networks, SAEs
JEPA, Flow Matching, GEPA
Conformal Prediction, Evals
RL: PPO, DPO, GRPO, DQN, MCTS
Data-Centric ML, Active Learning
Neural Data Science, LFP/RNA-seq

Systems & Domains
Physics, Neuroscience, Organoid Data
Signal Processing, CV, OCR/Layout
Physical AI, CAD/CAM, DFM
Process & Manufacturing Automation
Cognitive Security, AIxCC
Distributed Inference, FSDP, SLURM/HPC
Flash Attention, MCP, Agents
DSPy, Verifiers, Claude/Codex

Stack
Python, MATLAB, Nushell
TypeScript, SQL, FastAPI
Rust, Zig, C/C++, OCaml
PyTorch, PyG, Transformers
TensorFlow, OpenCV
Java, Lab Instrumentation
AWS, Docker, Linux

Endorsements

"Brings a novel and timely perspective to all projects." — Dr. Bob McGraw, CTO and Vice President, RAM Laboratories.
"Uncanny ability to bring abstract ideas to reality." — Brett Lewis, Senior Engineering Manager, Applied Medical.
"Mind of an inventor mixed with an artist." — Jonathan Mahan, Applied Medical.
"Students go farther because of Malachi's teaching." — Kayla Holman, Inspirit AI.